$\boxed{L21}$ e.g. $\mathcal{T}: \mathcal{L}_\infty[0,\infty) \longrightarrow \mathcal{L}_\infty[0,\infty)$ (Convolution) sup-norm

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$$\mathcal{T}(u(t)) := \int_0^t f(t-\tau)\, u(\tau)\, d\tau \quad , \text{ where } f(t) \in \mathcal{L}_1[0,\infty)$$

$$\|\mathcal{T}(u(t))\|_\infty = \left| \int_0^t f(t-\tau)\, u(\tau)\, d\tau \right| \leqslant \|f(t)\|_1 \, \|u(t)\|_\infty \quad \forall\, t \in \mathbb{R}$$

[Holder's inequality
$\left| \int_\Omega f(x) g(x) dx \right| \leqslant \|f(x)\|_p \, \|g(x)\|_q$
where $\frac{1}{p} + \frac{1}{q} = 1$]

$\square$ Let $\mathcal{H}_1$ and $\mathcal{H}_2$ be two Hilbert spaces, and let A be a bounded linear operator from $\mathcal{H}_1$ to $\mathcal{H}_2$. Then $\exists$ a unique bounded linear operator

$$A^*: \mathcal{H}_2 \longrightarrow \mathcal{H}_1$$

such that $\forall\, x \in \mathcal{H}_1$ and $y \in \mathcal{H}_2$, $\langle Ax, y \rangle = \langle x, A^* y \rangle$.

A^* is called an adjoint of A.

If $A^* = A$, then the operator is called self-adjoint operator.

e.g. $A: \mathbb{R}^n \longrightarrow \mathbb{R}^m$ $\quad A \in \mathbb{R}^{m \times n}$ $\quad$ Find: $A^*: \mathbb{R}^m \longrightarrow \mathbb{R}^n$

s.t. $\langle Ax, y \rangle = \langle x, A^* y \rangle \quad \forall\, x \in \mathbb{R}^n$, $y \in \mathbb{R}^m$

$$\underline{\langle Ax, y \rangle} = y^T(Ax) = (Ax)^T y = x^T A^T y = x^T (A^T y) = (A^T y)^T x = \underline{\langle x, A^T y \rangle}$$

$$A^* = A^T$$

e.g. $\mathcal{X} := \mathcal{L}_2[0,1]$ $\qquad \mathcal{Y} = \mathbb{C}^2$

$$A: \mathcal{X} \longrightarrow \mathcal{Y}$$

$$A(x(t)) = \begin{bmatrix} \int_0^1 t\, x(t)\, dt \\ \int_0^1 t^2 x(t)\, dt \end{bmatrix}$$

$x(t) \in \mathcal{L}_2[0,1]$

$$\langle Ax, y\rangle = y^* A x \qquad y := \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

$$= \begin{bmatrix} y_1^* & y_2^* \end{bmatrix} \begin{bmatrix} \int_0^1 t\,x(t)\,dt \\ \int_0^1 t^2 x(t)\,dt \end{bmatrix}$$

$$= y_1^* \int_0^1 t\,x(t)\,dt + y_2^* \int_0^1 t^2 x(t)\,dt = \int_0^1 \left[t\,y_1^* x(t) + t^2 y_2^* x(t) \right] dt$$

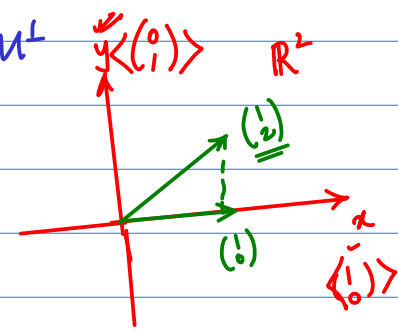
$$= \underbrace{\int_0^1 \left[t\,y_1(t) + t^2 y_2(t) \right]^* x(t)\,dt}_{\text{Inner product in } L_2[0,1]} = \langle x(t), A^* y(t) \rangle$$

$$\boxed{A^* y(t) = t\,y_1(t) + t^2 y_2(t)}$$

Let H be a Hilbert space and $M \subseteq H$ be a closed subspace. A bounded linear operator $P : H \to H$ is called the orthogonal projection onto M if

$$P(x+y) = x \quad \forall\, x \in M \text{ and } y \in M^\perp.$$

$Pv = x$ where $v \in H$ and $x \in M$ s.t. $v = x + y$, where $y \in M^\perp$



e.g. $P : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$

$\underline{P}\underline{x} = x_1$ where $x = x_1 + x_2$ $\quad x_1 \in \langle \binom{1}{0} \rangle$, $x_2 \in \langle \binom{0}{1} \rangle$

$\mathbb{R}^2$

$\binom{1}{0}, \binom{0}{1}$

$P\binom{1}{0} = \binom{1}{0}$ $\qquad P\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

$P\binom{0}{1} = \binom{0}{0}$ $\qquad \Rightarrow P = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

$M = \langle \binom{1}{1} \rangle$, $M^\perp = \langle \binom{1}{-1} \rangle$

What will be the operator for orthogonal projection onto $\langle \binom{1}{1} \rangle$

Let $S \subset \mathbb{C}$ be an open set and $f(s)$ be a complex valued function defined on S:

$$f(s): S \longrightarrow \mathbb{C}$$

Then $f(s)$ is said to be analytic at a point $z_0 \in S$ if it is differentiable at z_0.

* $f(s)$ is analytic at z_0 $\Longleftrightarrow$ $f(s)$ has continuous derivatives of all orders at z_0.

A function $f(s)$ is analytic on S if it is analytic at each point in S or if it has a derivative in S.

e.g. Stable transfer matrices are analytic in $\mathbb{C}^+$.

e^{-s} ?

$$\ell_p(\mathbb{J}),\ \ell_\infty(\mathbb{J}),\ \mathcal{L}_p(-\infty,\infty),\ \mathcal{L}_p[0,\infty),\ \mathcal{L}_p(-\infty,0]$$

Hilbert space $\rightarrow$ $\underbrace{\mathcal{L}_2,\ \mathcal{L}_{2+},\ \mathcal{L}_{2-}}$, $\underbrace{\ell_{2+},\ \ell_{2-},\ \ell_2}$